Fraud Risk Indicator Distribution

The FREQ Procedure

risk_zero_amount	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	282982	99.36	282982	99.36
1	1825	0.64	284807	100.00

risk_high_amount	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	275315	96.67	275315	96.67
1	9492	3.33	284807	100.00

risk_time_window	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	273213	95.93	273213	95.93
1	11594	4.07	284807	100.00

risk_flag	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	262190	92.06	262190	92.06
1	22617	7.94	284807	100.00

Combined Risk Flag vs Actual Fraud

The FREQ Procedure

Frequency

Table of risk_flag by fraud_flag

risk_flag	*fraud_flag* *0*	*1*	*Total*
0	261822	368	262190
1	22493	124	22617
Total	284315	492	284807

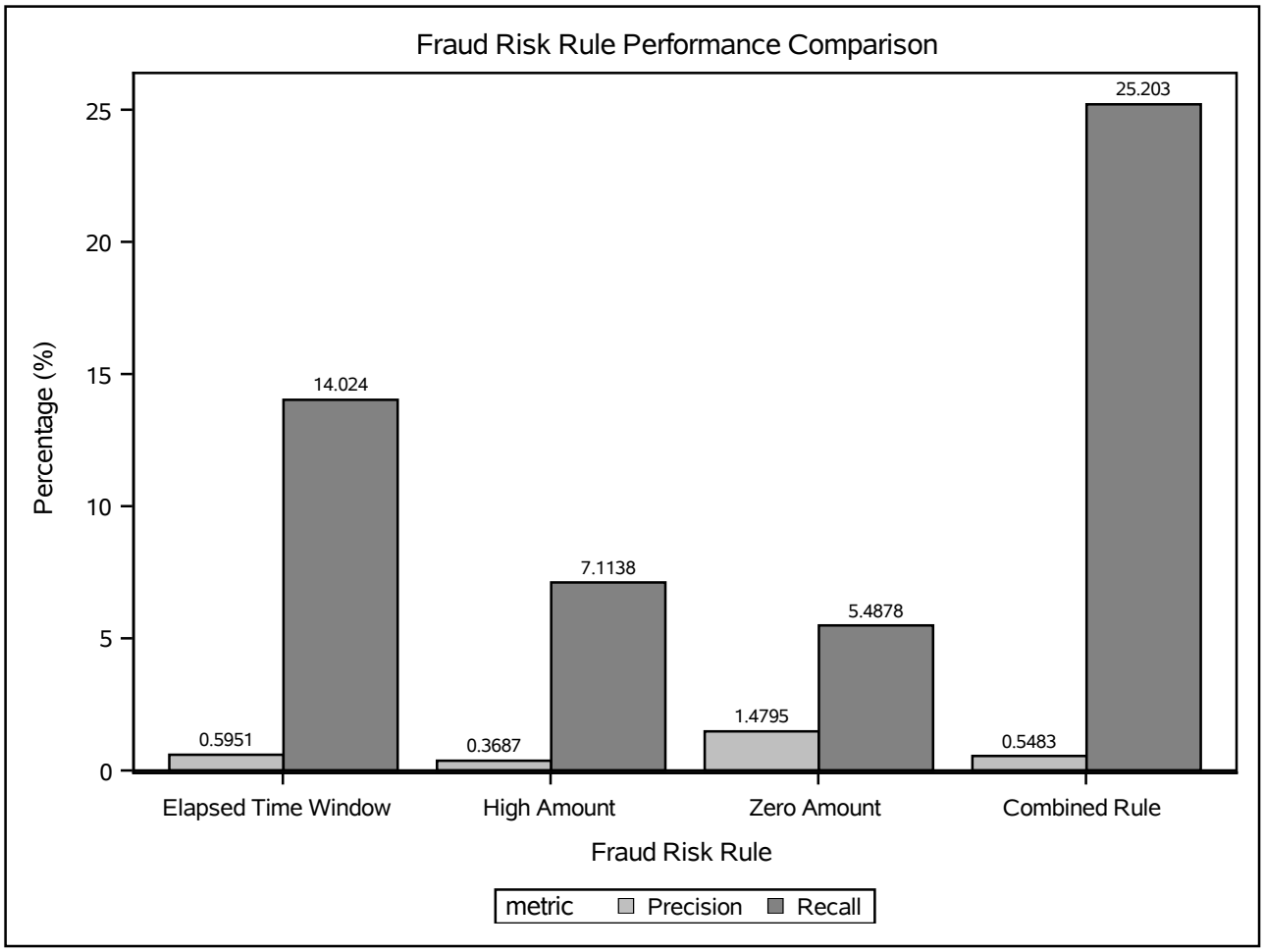
true_positive	*false_positive*	*false_negative*	*true_negative*
124	22493	368	261822

Combined Fraud Risk Indicator Performance

TP	FP	FN	TN	precision	recall	false_positive_rate	accuracy
124	22493	368	261822	0.55	25.20	7.91	91.97

Individual Fraud Risk Rule Comparison

rule_name	*total_alerts*	*TP*	*FP*	*FN*	*precision*	*recall*	*false_positive_rate*
Elapsed Time Window	11594	69	11525	423	0.60	14.02	4.05
High Amount	9492	35	9457	457	0.37	7.11	3.33
Zero Amount	1825	27	1798	465	1.48	5.49	0.63


Fraud Risk Rule Performance Comparison
Percentage (%)
0
5
10
15
20
25
0.5951
14.024
0.3687
7.1138
1.4795
5.4878
0.5483
25.203
Elapsed Time Window
High Amount
Zero Amount
Combined Rule
Fraud Risk Rule
metric
Precision
Recall